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Quality Inspection and Defect Recognition Technology for Intelligent Manufacturing Based on Machine Vision

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***Abstract*—With the rapid development of intelligent manufacturing, quality inspection and defect recognition have become one of the key technologies to improve production efficiency and product quality. Traditional manual inspection methods cannot meet the demands of efficient, accurate and real-time production, so the quality inspection and defect recognition technology based on machine vision has gradually become an important application in the field of intelligent manufacturing. This paper firstly introduces the basic principles of machine vision technology and the background of its application in intelligent manufacturing, and then discusses in detail the key technologies in quality inspection and defect recognition, including image acquisition and pre-processing, feature extraction and matching algorithms, as well as defect classification and determination methods. This paper also analyzes the design and implementation of the intelligent manufacturing quality inspection system, including the system architecture, data processing flow and practical application cases. An outlook is given for the challenges faced by the current technology and the future development trend. This paper aims to provide reference for researchers and engineers in related fields to promote the further development and application of machine vision technology in intelligent manufacturing.**

Keywords—Machine Vision, Intelligent Manufacturing, Quality Inspection, Defect Recognition, Image Processing, Feature Extraction, Deep Learning, System Design

I. Introduction

With the acceleration of the global manufacturing industry's transformation to intelligence and digitization, intelligent manufacturing has become an important way to improve productivity, optimize product quality, and reduce costs[1]. In this process, quality inspection and defect recognition technology plays a crucial role as a key link to ensure product quality[2]. The traditional manual inspection method is limited by the inspector's experience and subjective judgment, which is difficult to meet the high requirements of precision, efficiency and real-time in mass production, and machine vision technology, as an automated, efficient and accurate means of inspection, has gained wide attention and application[3].

Machine vision technology can realize rapid inspection and defect identification of products by simulating the human visual system and collecting, processing and analyzing images in the production process[4]. In intelligent manufacturing, the machine vision system can not only improve the accuracy of detection, but also enhance the automation level of the production line and reduce human error. Especially in the field of high precision requirements, such as electronic products, automobile manufacturing and precision instruments, machine vision technology demonstrates its great application potential[5].

The purpose of this paper is to discuss the latest progress of machine vision-based quality inspection and defect recognition technology, analyze its application value in intelligent manufacturing, and study the technical challenges it faces[6]. The basic principles of machine vision and its application background in intelligent manufacturing are introduced, followed by a detailed discussion of the key technologies in quality inspection and defect recognition, including image acquisition, feature extraction, and defect classification. Combined with practical application cases, we analyze the realization and application effect of machine vision technology in intelligent manufacturing, and look forward to its future development trend.

II. Overview of Machine Vision Application in Intelligent Manufacturing

Machine vision technology is to simulate the human visual system, use image acquisition equipment (such as cameras, sensors, etc.) to obtain image data in the production process, and process, analyze and understand the image through computer vision algorithms, so as to realize the automatic detection, measurement and identification of objects. As an important part of the automated manufacturing process, machine vision technology can effectively replace manual visual inspection and improve the degree of automation, inspection efficiency and accuracy of the production line. Image Preprocessing (Noise Reduction):

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (1)$$

Smart manufacturing emphasizes an efficient, flexible and precise production process, of which quality inspection is an integral part[7]. In traditional manufacturing, quality inspection often relies on manual operation, facing the problems of high labor cost, low efficiency and high error[8]. Machine vision system can carry out real-time and accurate quality inspection in every link of production, especially in batch and refined production, which can significantly improve the qualification rate of products, reduce the defect rate, and guarantee the consistency and high quality of products.

Machine vision technology through high-resolution camera equipment and advanced image processing algorithms, can realize the product surface, internal structure and other aspects of defect recognition[9]. Whether it is tiny cracks, scratches, or size deviation, machine vision system can be accurately identified through image analysis[10]. In the

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process of intelligent manufacturing, machine vision is widely used in defect detection in electronic products, automobile manufacturing, metal processing and other industries, which can realize efficient control of product quality without affecting production speed, showed in Figure 1:

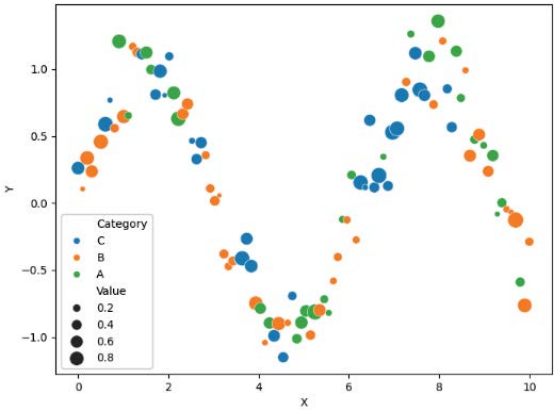

Figure 1 Scatterplot with Hue and Size

The application of machine vision in intelligent manufacturing has many advantages, it can achieve efficient, stable and continuous automated quality inspection; machine vision system can improve the accuracy and sensitivity of detection by continuously optimizing the algorithm[11]. Although machine vision technology shows great potential in quality inspection, environmental changes in image acquisition, target detection in complex backgrounds, real-time and stability of algorithms and other issues[12]. With the continuous progress of computer vision and deep learning technology, these challenges are expected to be effectively solved in the future, thus promoting the wide application of machine vision in the field of intelligent manufacturing. Feature Extraction (Edge Detection using Sobel Operator):

$$G_x = \left[\!\left[\{bmatrix\} -1 \,\&\, 0 \,\&\, 1 \backslash\backslash -2 \,\&\, 0 \,\&\, 2 \backslash\backslash -1 \,\&\, 0 \,\&\, 1 \right]\!\right] \{bmatrix\} \quad (2)$$

III. Key Technologies for Quality Inspection and Defect Recognition

The key technologies for quality inspection and defect recognition mainly include image acquisition and preprocessing, feature extraction and matching algorithms, and defect classification and determination technologies[13]. These technologies work together to form a complete quality inspection system. Image acquisition and preprocessing technology ensures that the acquired image data is clear and accurate through efficient image acquisition equipment and processing methods[14]; feature extraction and matching algorithms are responsible for extracting useful information from the image and comparing it with the standard model to identify potential defects; defect classification and determination technology further classifies and determines the recognized defects through deep learning or traditional algorithms, and finally The defect classification and determination technology further classifies and determines the recognized defects through deep learning or traditional algorithms, and finally evaluates the product quality[15]. The above three technologies constitute a comprehensive and efficient quality inspection and defect recognition system, which is the core of high-precision inspection in intelligent manufacturing[16].

A. Image Acquisition and Preprocessing Technology

Image Acquisition Equipment and Technology

The core of image acquisition equipment is the camera. According to the demand, the commonly used camera types include line scan cameras, surface array cameras and industrial cameras. Line scan cameras are suitable for inspection of long targets or high speed production lines, while face array cameras are widely used for image capture in static or low speed production. Industrial cameras, compared with ordinary consumer-grade cameras, have higher resolution, lower noise and stronger anti-jamming ability to meet the needs of high-precision image acquisition in complex environments, showed in Figure 2:

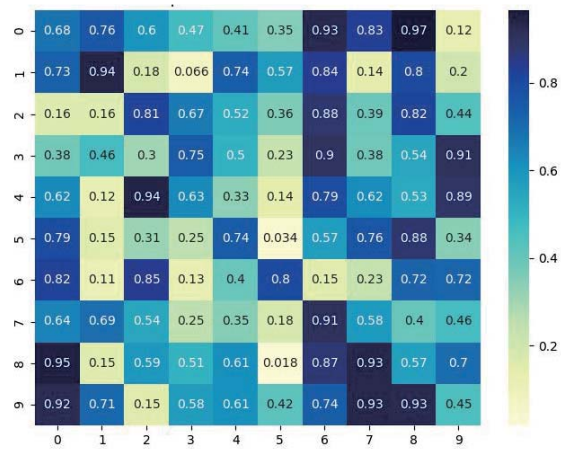

Figure 2 Heatmap of Random Correlation Matrix

The choice of light source is crucial to the image capture effect. Due to the reflective properties of the target object, surface morphology and changes in the surrounding environment, the type and layout of the light source will directly affect the quality of the image[17]. Common types of light sources include LED light sources, laser light sources and halogen lamps. LED light sources are widely used in various types of machine vision systems due to their stability and adjustability, while laser light sources are commonly used for the detection of objects with low contrast or tiny defects.

The choice of lens is another key factor that affects the quality of imaging. The main function of the lens is to focus the object and project the image onto the camera's sensor. A high quality lens ensures image clarity and accuracy, especially when detecting fine defects. Parameters such as focal length, aperture, field of view, and aberration need to be considered when choosing the right lens to ensure that image details are accurately captured.Image acquisition technology includes image acquisition cards, interface standards and data transmission methods[18]. Commonly used image acquisition cards support a variety of interface protocols, USB, GigE, Camera Link and CoaXPress, etc., enabling high-speed image data transmission[19]. With the development of industrial Ethernet and wireless technology, real-time transmission and remote processing of image data also become possible, further enhancing the application flexibility and adaptability of machine vision systems. Support Vector Machine (SVM) Decision Function:

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$$f(x) = w^T x + b \quad (3)$$

B. Feature Extraction and Matching Algorithms

Feature Extraction Methods

Edge-based feature extraction methods are the most traditional and widely used class of techniques. Edge detection is mainly used to extract the contour and shape information of objects by identifying regions in the image where the gray scale changes drastically[20]. Common edge detection algorithms are Canny edge detection, Sobel operator and Laplacian operator. These methods can effectively extract the boundary information of the object. For images with complex backgrounds or more noise, edge extraction may be affected and additional image preprocessing steps are required to improve the accuracy.

Corner point based feature extraction methods are usually used in regions of an image that are stable and have strong localized features[21]. Corner points are points in an image that have large gray scale variations and high stability, Harris corner point detection algorithm and Shi-Tomasi corner point detection algorithm[22]. The corner point feature extraction method is highly resistant to interference during the matching process, especially when dealing with rotations, scaling and changes in viewing angle, and can provide more stable recognition results. This method is commonly used for object recognition and 3D reconstruction in scenes.

Texture-based feature extraction methods, on the other hand, focus on the texture information of local regions in the image, such as roughness and smoothness. Commonly used texture-based feature extraction methods include gray level covariance matrix (GLCM), local binary pattern (LBP) and Gabor filter. These methods are able to identify small structural differences on the surface by analyzing the local texture features of the image, and are widely used in surface defect detection and quality control. Compared with edge and corner point methods, texture features can better accommodate complex textured backgrounds and low-contrast objects. Convolutional Neural Network (CNN) Convolution Operation:

$$(I * K)(x, y) = \sum_m \sum_n I(x + m, y + n) K(m, n) \quad (4)$$

Feature extraction methods based on deep learning have gained widespread attention in recent years, and have especially demonstrated strong capabilities in processing complex images. Deep learning automatically learns representative features from the original image by constructing a multi-layer neural network, which usually can effectively represent multi-dimensional information such as shape, texture, and color of an object. Common deep learning methods include convolutional neural networks (CNN), regional convolutional neural networks (R-CNN), etc. These methods have achieved significant results in tasks such as image classification, target detection and defect recognition, and deep learning methods have stronger adaptability to handle more complex and variable image data, showed in Figure 3:

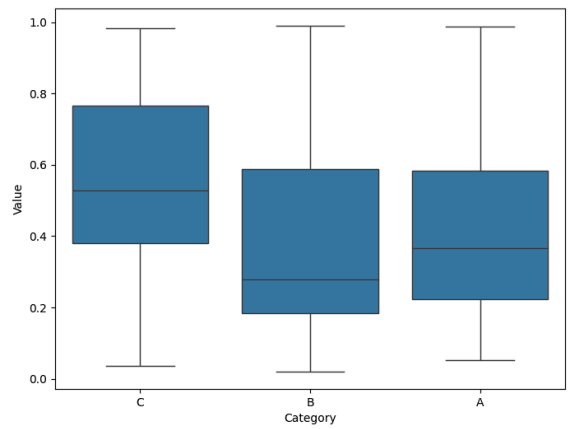

Figure 3 Boxplot for Distribution Comparison by Category

C. Defect Classification and Determination Techniques

Defect Classification Methods Based on Machine Learning and Deep Learning

Support Vector Machine (SVM) is a supervised learning method commonly used for defect classification, which is especially suitable for solving the binary classification problem. SVM distinguishes data points of different classes by finding the optimal decision boundary. In defect detection, SVM is able to categorize normal and defective products based on the features extracted from the image. The advantage of SVM is its better ability to handle small-sample data and its high classification accuracy, which makes it especially suitable for those situations where the number of defective samples is relatively small. When facing large-scale datasets, the computational complexity of SVM is high, which may lead to inefficient processing.

Decision trees are an intuitive and easy-to-understand classification method commonly used in defect determination. Decision trees construct a classification model by recursively selecting the optimal features, thus realizing the effective differentiation of different categories. In defect recognition, the decision tree can determine whether there is a defect based on image features hierarchically, which is suitable for defect categories with obvious distinguishing features. Although the decision tree model is easy to implement and the results are easy to interpret, it may suffer from overfitting, especially when the dataset is not rich enough, and appropriate pruning operations are needed to improve the generalization ability. Decision Tree Split Criterion (Information Gain):

$$IG(D, A) = Entropy(D) - \sum_{v \in \text{Values}(A)} \frac{|D_v|}{|D|} \cdot Entropy(D_v) \quad (5)$$

Convolutional neural network (CNN) is one of the most commonly used algorithms in deep learning for image processing and defect classification. CNN is able to process complex image data and perform accurate defect classification by automatically extracting higher-order features from images through multilayer convolutional operations. In defect recognition, CNN can effectively capture local patterns and global structures in images without manually designing features, thus automatically learning features suitable for defect recognition. Especially in the case of complex backgrounds, tiny defects, or multiple defect types, CNNs show excellent performance. Although deep learning models such as CNNs offer significant advantages in terms of accuracy and degree of automation, they also require large amounts of training data and efficient computational

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resources, which can be a challenge in some industrial applications.

Extended versions of convolutional neural networks, such as Regional Convolutional Neural Networks (R-CNN) and YOLO (You Only Look Once), are also widely used in defect classification and determination. R-CNN enables more accurate defect localization and classification by segmenting an image into regions and performing convolutional analysis on each region. YOLO, on the other hand, enables more accurate defect localization and classification by YOLO achieves real-time and efficient defect detection by simultaneously localizing and classifying objects in a single network. The advantages of these methods in defect recognition are their efficiency and accuracy, which are especially suitable for real-time monitoring and defect classification applications in batch production lines, showed in Figure 4:

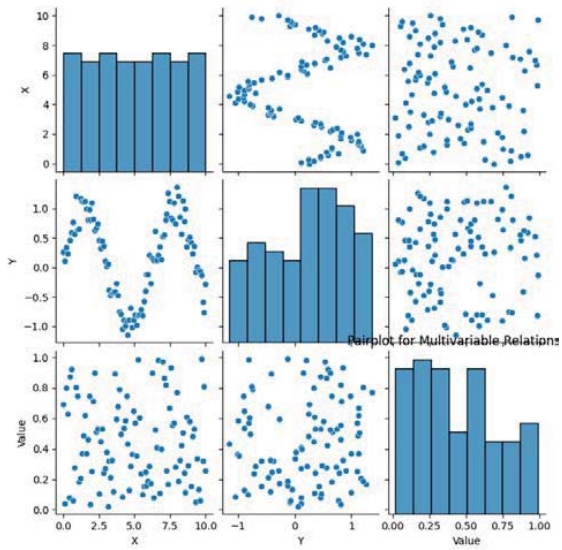

Figure 4 Pairplot for Multivariable Relationships

IV. Design and Implementation of Intelligent Manufacturing Quality Inspection System

Hardware configuration is the foundation of an intelligent manufacturing quality inspection system. The system usually consists of an image acquisition device (e.g., an industrial camera), a lighting system, an image processing unit, and an actuation unit. The industrial camera is responsible for collecting image data of the product, and the lighting system ensures the clarity and contrast of the image by adjusting the angle and intensity of the light source. The image processing unit is the core part of the system, which is responsible for processing, feature extraction and defect analysis of the collected images. The actuator unit provides automated feedback based on the inspection results, such as product categorization, defect marking or production line adjustment. Therefore, the rationality of hardware configuration directly affects the performance and stability of the system.

The design of image processing and analysis algorithms is the key technology in the quality inspection system. The system needs to pre-process the image data, including operations such as denoising, enhancement and edge detection, to ensure the accuracy of subsequent analysis. The feature extraction and matching algorithm then extracts meaningful features from the image, such as edges, corners, textures, etc., and carries out defect classification and identification through machine learning or deep learning algorithms. The choice of algorithms needs to be decided according to the actual application scenarios, such as for the detection of tiny defects, which may require the use of high-resolution cameras and more detailed image processing algorithms. In addition, the real-time performance of the algorithms is also an important consideration in the design, which must ensure that large amounts of data can be processed quickly on the production line. Classification Accuracy:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (6)$$

The design of the data transmission and processing architecture determines the responsiveness and scalability of the system. Modern intelligent manufacturing quality inspection systems usually use a distributed architecture to transmit data from the collection end to the processing end through industrial Ethernet or wireless communication technology. This process needs to ensure the stability and security of data transmission. The data center or cloud can process and store the inspection data in real time for subsequent quality analysis and decision support. In addition, the introduction of edge computing technology can enable part of the data processing and decision-making process to be completed at the device end, thus reducing the dependence on the central server and improving the response speed and flexibility of the system.

The implementation and optimization of the intelligent manufacturing quality inspection system is the last link in the system design. In the process of system realization, the system needs to be fully tested and debugged to ensure that it can operate stably under various working conditions. As the production environment changes and new defect types appear, the system needs to be continuously optimized and upgraded. For example, by introducing incremental learning and online learning algorithms, the system can continuously update and improve the defect recognition capability during operation. In addition, the accumulation of user feedback and production data also contributes to the continuous optimization of the system, making the intelligent manufacturing quality inspection system increasingly intelligent and automated in actual production, and ultimately achieving the goal of efficient and accurate quality control, showed in Figure 5:

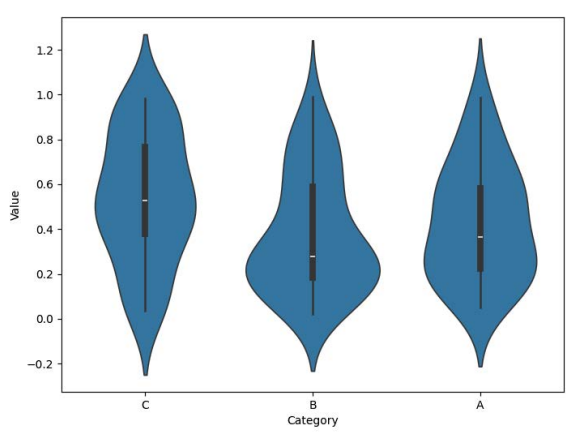

Figure 5 Violin Plot for Detailed Distribution

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V. CONCLUSION

Intelligent manufacturing quality inspection and defect recognition technology has an important application value in modern manufacturing industry, especially plays a key role in improving production efficiency, reducing costs and guaranteeing product quality. By analyzing the application of machine vision technology in intelligent manufacturing, this paper discusses the key technologies of image acquisition and preprocessing, feature extraction and matching, defect classification and determination, and demonstrates the advantages of innovative methods based on machine learning and deep learning in quality inspection.

With the continuous development of technology, the intelligent manufacturing quality inspection system has been transformed from traditional manual inspection to automation and intelligence, and has become a core tool to improve the level of industrial production. The introduction of advanced technologies such as deep learning algorithms, edge computing, cloud computing, etc., has greatly contributed to the improvement of system performance, making defect identification more accurate and real-time. In practical application, the intelligent quality inspection system can quickly respond to the needs of the production line, monitor and detect defects in real time, and ensure the consistency and reliability of product quality. The current intelligent manufacturing quality inspection system has achieved remarkable results in some areas, and with the further development of artificial intelligence technology, sensing technology and data processing capabilities, the intelligent manufacturing quality inspection system will be more intelligent, automatic and efficient, and provide a more solid technical guarantee for the high-quality development of the manufacturing industry.

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